

MYILVAGANAN SAKTHIVEL

Senior Software Engineer | Full-Stack & AI-Integrated Systems | Team Lead

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PROFESSIONAL SUMMARY

Results-driven Senior Software Engineer and Technical Lead with 7+ years of experience architecting and delivering production-grade full-stack applications using the MERN stack (MongoDB, Express.js, React.js, Node.js), Python, and TypeScript. Proven track record in leading cross-functional engineering teams, driving end-to-end delivery from requirements to production, and mentoring developers to uphold SOLID principles and clean architecture standards.

Deep expertise in cloud-native, event-driven architectures across AWS (Lambda, DynamoDB, API Gateway, CDK) and Azure (IoT Hub, Event Hub, Service Bus, Cosmos DB, WebPubSub). Hands-on experience integrating AI/LLM-powered agentic workflows, RAG pipelines, and intelligent automation into production systems. Skilled in real-time IoT telemetry processing, scalable API design, performance optimisation, CI/CD pipelines, and production incident management. Actively upskilling in AI-integrated engineering and intelligent monitoring for enterprise platforms.

CORE COMPETENCIES

Technical Leadership · System Design & Architecture · MERN Stack Development · Python Backend Engineering
AI/LLM Integration · Agentic Workflow Automation · RAG Pipelines · Prompt Engineering · Event-Driven Architecture
REST API Design · Microservices · Cloud-Native Development (AWS & Azure) · IoT Edge-to-Cloud · Real-Time Systems
Agile / Scrum · Stakeholder Management · Code Review & Mentoring · CI/CD Pipelines · Performance Optimisation

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python

Frontend: React.js, Next.js, Redux, Redux-Saga, HTML5, CSS3, AG Grid, Stencil UI

Backend: Node.js, Express.js, NestJS, Python (FastAPI, Flask), REST APIs, Microservices

Databases: MongoDB, MySQL, DynamoDB, Cosmos DB, Redis

AI / ML: LangChain, OpenAI API, Prompt Engineering, RAG Pipelines, Agentic Workflows, LLM Integration, AI Automation

Cloud – AWS: Lambda, API Gateway, DynamoDB, S3, CloudFront, AppConfig, CDK, AWS SDK

Cloud – Azure: App Services, IoT Hub, Event Hub, Service Bus, Blob Storage, WebPubSub, Cosmos DB, Redis

DevOps: GitLab CI/CD, Jenkins, AWS CodePipeline, Docker (concepts), Kubernetes (concepts)

Testing: Jest, Cypress, Pytest, Unit Testing, Integration Testing

Monitoring: Kibana, QuickSight, Application Insights, Azure Monitor

Tools: Git, GitHub, NPM, Yarn, Webpack, Babel, Figma, Jira, Confluence

Practices: SOLID Principles, Design Patterns, OOP, Clean Architecture, TDD, Code Reviews, Agile/Scrum

PROFESSIONAL EXPERIENCE

CGI Information Systems and Management Consultants Pvt. Ltd.

Senior Software Engineer (Technical Lead) · Feb 2025 – Present · Bangalore

Client: CircleK, United States

- Spearheaded end-to-end architecture and delivery of a retail proactive monitoring platform for fuel operations across 500+ retail sites in the US, improving system uptime and reducing operational downtime through automated health tracking and real-time dashboards.
- Designed and built scalable Node.js REST APIs and Python backend services powering React.js dashboards for device status monitoring, deployment workflows, event tracking, and operational insights — reducing average investigation time by ~45%.
- Architected a high-throughput event-driven pipeline using Azure Service Bus + WebPubSub processing 100K+ device events/day, enabling near real-time UI updates and scalable fan-out to multiple dashboard consumers.
- Built and led delivery of an AI-assisted agentic automation system using LLM-based log parsing and intelligent rule evaluation, replacing manual log analysis workflows and reducing manual operational effort by ~60%.
- Engineered a Python IoT telemetry pipeline for edge-to-cloud ingestion, processing and normalising device events from fuel pump sensors across geographically distributed sites with consistent validation and routing.
- Designed an EV alerting engine with configurable threshold rules from UI, enabling real-time rule evaluation and alerting for EV charger health — reducing false-positive alerts through intelligent condition chaining.
- Optimised bulk data retrieval APIs with streaming support and efficient pagination strategies, reducing dashboard load time under large concurrent workloads by ~40%.
- Automated end-to-end deployment workflows by integrating deployment APIs with Ansible execution and live status streaming via messaging queues — eliminating manual deployment tracking.
- Mentored and guided junior and mid-level engineers on MERN stack best practices, SOLID principles, clean architecture, and design patterns through structured code reviews and design walkthroughs.
- Collaborated directly with US-based client stakeholders (CircleK) to translate business requirements into technical roadmaps, define sprint milestones, and ensure on-time delivery of platform features.

Infosys Ltd.

Consultant (Technical Lead) · Sep 2021 – Feb 2025 · Bangalore

Client: Amazon India

- Led a cross-functional team of 5–7 developers across multiple concurrent modules on Amazon India's Rewards and Campaign Management platform, serving millions of users.
- Architected and delivered the Rewards Central Application Externalisation using React.js (TypeScript), Redux, AG Grid, and Spring Boot APIs — reducing campaign creation cycle time by ~35% through workflow automation and optimised data grids.
- Designed and implemented serverless AWS Lambda microservices integrated with API Gateway, handling over 1M+ API calls/month for campaign data operations with consistent sub-200ms response times.
- Built the Nudge Tool — an email marketing automation system using AWS DynamoDB and Stencil UI — enabling targeted customer segmentation and reducing campaign targeting turnaround from 3 days to under 4 hours.
- Developed the Bulk Setup Module for mass campaign creation, integrating CKEditor for rich content editing and AWS AppConfig-backed CSV validation — supporting ingestion of 10K+ campaign records in a single operation.
- Built the Admin Console Module for advertiser onboarding using React.js, Redux, and MySQL, streamlining approval workflows for 200+ active advertisers and reducing onboarding time from weeks to days.
- Led the Rewards to Funder Console Migration, migrating critical business logic and adding real-time performance monitoring dashboards using Kibana and QuickSight for campaign analytics.
- Built React.js + Redux-Saga modules with memoisation and virtualised rendering for high-traffic e-commerce pages, improving page load performance by ~30%.
- Implemented Python and Java services for payment processing integrations, ensuring transactional accuracy and fault tolerance with retry and circuit-breaker patterns.

- Ensured WCAG 2.1 AA accessibility compliance across all UI modules and drove cross-browser optimisation, expanding platform reach across diverse user groups.
- Conducted technical interviews, onboarded new team members, led Agile sprint ceremonies (planning, stand-ups, retrospectives), and served as the primary technical point of contact for product and business stakeholders.
- Received Engineering Excellence Award (2022) and Best Enabler Award (2023) for outstanding technical leadership and delivery contributions.

Straive (previously SPi Global Information Technology Ltd.)

Software Developer · Jul 2021 – Aug 2021 · Chennai

- Designed and developed an internal remote-access file directory system using React.js, JavaScript, Bootstrap, and CSS — simplifying navigation for internal teams and reducing file-access friction.
- Built a file preview feature allowing users to inspect files before downloading or deleting, improving operational efficiency and reducing accidental deletions.
- Collaborated with stakeholders through iterative UX design feedback cycles, producing UI wireframes from scratch and refining based on usability testing outcomes.
- Documented design decisions and maintained project continuity artifacts for future team onboarding.

MothersonSumi Infotech and Designs Ltd.

Engineer · Aug 2018 – Jan 2021 · Noida

- Built dynamic web applications using JavaScript (ES6+), DOM manipulation, HTML5, CSS3, AJAX, Webpack, and Babel, laying a strong full-stack foundation.
- Contributed to UI/UX design processes using Figma and Adobe XD, collaborating with design and product teams to translate wireframes into functional interfaces.
- Actively diagnosed and resolved code-level bugs and performance bottlenecks, improving delivery timelines through systematic troubleshooting.
- Participated in on-site client assignments, gaining hands-on exposure to enterprise delivery practices and cross-team collaboration workflows.

DEVELOPER PROJECTS

DevConnector – Developer Social Platform

- Built a full-stack social networking platform for developers using the MERN stack (MongoDB, Express.js, React.js, Node.js), enabling profile creation, GitHub integration, and developer community features.
- Implemented JWT-based authentication, profile CRUD, post creation with likes/comments, and real-time feed; deployed on Heroku with environment-based configuration.
- Tech: MongoDB, Express.js, React.js, Node.js, Redux, JWT, Heroku | [GitHub](#)

Green Events – Environmental Event Platform (Next.js)

- Developed a Next.js SSR platform for posting and discovering environmental community events, using Strapi CMS as the headless backend with dynamic content management.
- Implemented server-side rendering for SEO optimisation, dynamic routing, and image optimisation; deployed on Vercel with CI integration.
- Tech: React.js, Next.js, Node.js, Strapi (Headless CMS), Vercel | [GitHub](#)

E-Commerce Flora Shop – MERN Full-Stack App

- Built a full-featured e-commerce platform for buying and selling plants using MERN stack, featuring product listings, cart management, user authentication, and order tracking.
- Designed RESTful APIs with Express.js, implemented MongoDB schemas for products/orders/users, and built an intuitive React.js storefront with Redux state management.
- Tech: MongoDB, Express.js, React.js, Node.js, Redux, Stripe (payment), Heroku | [GitHub](#)

Netflix Clone – React.js Streaming UI

- Developed a pixel-accurate Netflix UI clone using React.js with The Movie Database (TMDB) API integration, featuring dynamic movie categories, banner trailers, and responsive layout.
- Tech: React.js, TMDB API, CSS3 | [GitHub](#)

EDUCATION

Bachelor of Engineering – Mechatronics | CGPA: 8.98 | 2014 – 2018

K. S. Rangasamy College of Technology, Tiruchengode, Tamil Nadu

Higher Secondary Certificate (HSC) | 90.92% | 2013 – 2014

Kongu Matriculation Higher Secondary School, Salem, Tamil Nadu

Secondary School Leaving Certificate (SSLC) | 96% | 2011 – 2012

Kongu Matriculation Higher Secondary School, Salem, Tamil Nadu

CERTIFICATIONS & AWARDS

- AWS Certified Developer – Associate
- MERN Stack Certification – Udemy
- AWS Cloud Essentials – CGI, 2025
- Introduction to Prompt Engineering – CGI, 2025
- Deep Learning and Neural Networks – CGI, 2025
- OS Processes and Threads – CGI, 2025
- CGI APAC: Linux – 2025
- Deploying a Python Script with Flask – CGI, 2025
- Infosys Quality Process Certification
- Engineering Excellence Award – Infosys, 2022
- Best Enabler Award – Infosys, 2023
- Silver Award for Exceptional Work – CGI, 2025